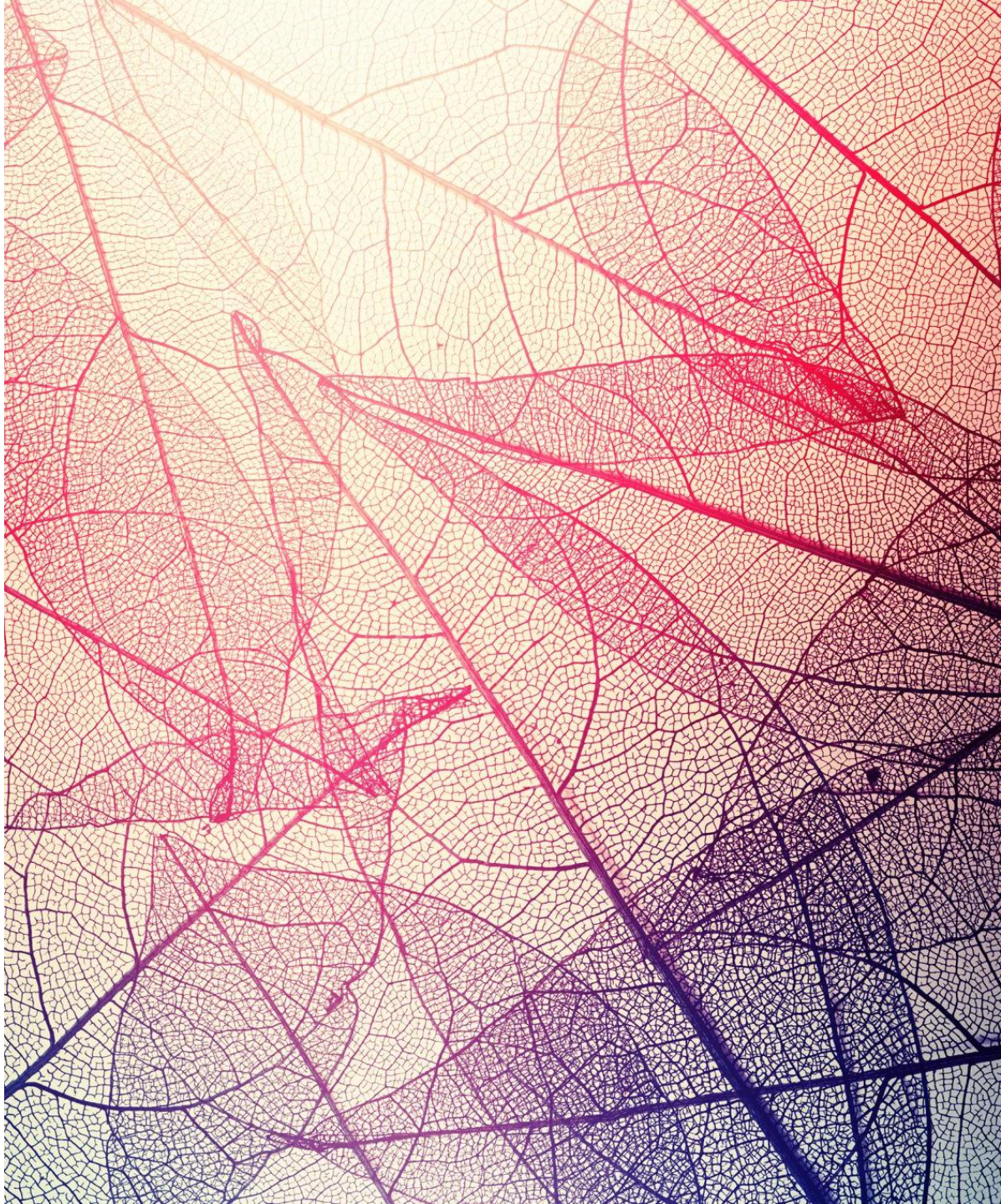
BRUCELLOSIS

Presented by: Haley Gebhard, Ifeoluwanimi Shobayo, and Chelsea Mazonde





DEFINITION OF PROBLEM

- Brucellosis: undulant fever or Mediterranean fever[1]
  - Affects humans and animals, especially cattle, dogs, sheep, goats[2]
  - Higher global incidence than previously thought: 1.6-2.1 million new human cases per year[2]
- Endemic in: Middle East, Mediterranean, Central and South America[2]
- Areas of high incidence rates, especially with few resources: Mediterranean, Middle East, Central Asia, certain areas in Africa[2]
  - Especially: Iran, Kazakhstan, Azerbaijan, Turkmenistan, Kyrgyzstan, Armenia, Tajikistan, and Uzbekistan
  - Central and South America: Mexico and Peru

1. California Department of Public Health. Brucellosis (Undulant fever, Mediterranean fever). Brucellosis Fact Sheet. September 2021. Accessed April 8, 2025. https://www.cdph.ca.gov/Programs/CID/DCDC/CDPH%20Document%20Library/BrucellosisFactSheet.pdf
2. Qureshi KA, Parvez A, Fahmy NA, et al. Brucellosis: epidemiology, pathogenesis, diagnosis and treatment-a comprehensive review. *Ann Med*. 2023;55(2):2295398. doi:10.1080/07853890.2023.2295398

DEFINITION OF THE PROBLEM CONT'D

Europe: annual incidence rate 0.09 per 100,000 people from 2017-2018[2]

- Decline in cases from 735 in 2008 to 352 in 2011[2]
- 2022: 29 countries reported brucellosis data, 18 of them with confirmed cases, totaling 199; 11 countries had 0 reported cases[3]

U.S.: California – 492 cases from 1993-2017, older Latino men consuming unpasteurized Mexican-style soft cheese[2]

Priority populations: older, Latino – especially from Mexico, consuming unpasteurized dairy products[2]

2. Qureshi KA, Parvez A, Fahmy NA, et al. Brucellosis: epidemiology, pathogenesis, diagnosis and treatment-a comprehensive review. *Ann Med.* 2023;55(2):2295398. doi:10.1080/07853890.2023.2295398
3. European Centre for Disease Prevention and Control. Brucellosis: Annual epidemiological report for 2022. Surveillance Report. April 2024. Accessed April 9, 2025. https://www.ecdc.europa.eu/sites/default/files/documents/BRUC_AER_2022_Report.pdf

BRIEF HISTORY OF BRUCELLOSIS

- History: found vertebral brucellosis in a skeletal remains of Pliocene *Australopithecus africanus*[4]
  - Dated at 2.3-2.5 million years old
- First recorded case: human skeleton from the Middle Bronze Age (about 3300 B.C.)
- Hippocrates (450 B.C.) description of brucellosis-like disease[5]
- Roman empire: cases of brucellosis increased, most likely endemic[4]
- Middle Ages: brucellosis likely endemic
- Found most often in adult male human skeletons, specifically in the lumbar vertebrae and sacroiliac joints
- First identified by David Bruce in 1887

4. D'anastasio R, Staniscia T, Milia M, et al. Origin, evolution and paleoepidemiology of brucellosis. *Epidemiol. & Infect.* 2010;139(1):149-156. doi:10.1017/S095026881000097X
5. Akpinar O. Historical perspective of brucellosis: a microbiological and epidemiological overview. *Infez Med*. 2016;24(1):77-86.

DESCRIPTION OF AGENT AND TRANSMISSION

- Brucella bacteria: infection from contact with infected animals, uncooked meat and raw milk products[6]
  - Main reservoirs: cows, feral swine, pigs, and dogs
- Four species and distribution of disease in the U.S. by species:
  - *B. melitensis* and *B. abortus* from eating unpasteurized dairy products; 70-75% of brucellosis cases; examples: raw milk and cheese
    - *B. abortus* strain, RB51, from *Brucella* vaccine for livestock; veterinary needle prick or drinking raw milk
  - *B. suis* from contact with feral swine, diseased carcass, or infected dog; 25-30% of brucellosis cases
  - *B. canis* from dogs; rare; causes illness in people but not severe

6. Centers for Disease Control and Prevention. Clinical overview of brucellosis. Brucellosis. May 2, 2024. Accessed April 8, 2025. https://www.cdc.gov/brucellosis/hcp/clinical-overview/index.html



CASE DEFINITION, INCUBATION PERIOD, DURATION

- Case definition and clinical description: acute onset of fever and one or more other symptoms[7]
  - Other symptoms: headache, night sweats; arthralgia; fatigue; myalgia; anorexia; weight loss; arthritis/spondylitis; meningitis; focal organ involvement, specifically orchitits/epididymitis, hepatomegaly, endocarditis, and splenomegaly[6,7]
- Laboratory Criteria:
  - Culture of Brucella spp. or acute- or convalescent-phase serum specimens[7]
  - Things to pay attention to: possible route of exposure in workplace, eating habits, traveling, hunting, veterinary work, laboratory procedures[6]
- Incubation period: 2-4 weeks; range 5 days-6 months[8]
- Duration: few weeks to several months[6]

6. Centers for Disease Control and Prevention. Clinical overview of brucellosis. Brucellosis. May 2, 2024. Accessed April 8, 2025. https://www.cdc.gov/brucellosis/hcp/clinical-overview/index.html
7. Centers for Disease Control and Prevention. Brucellosis (*Brucella* spp.) 2010 case definition. National Notifiable Diseases Surveillance System (NNDSS). April 16, 2021. Accessed April 8, 2025. https://ndc.services.cdc.gov/case-definitions/brucellosis-2010/
8. Centers for Disease Control and Prevention. Brucellosis: CDC Yellow Book 2024 Travel-Associated Infections & Diseases. Traveler's Health. May 1, 2023. Accessed April 18, 2025. https://wwwnc.cdc.gov/travel/yellowbook/2024/infections-diseases/brucellosis

OTHER TRANSMISSION PATHWAYS

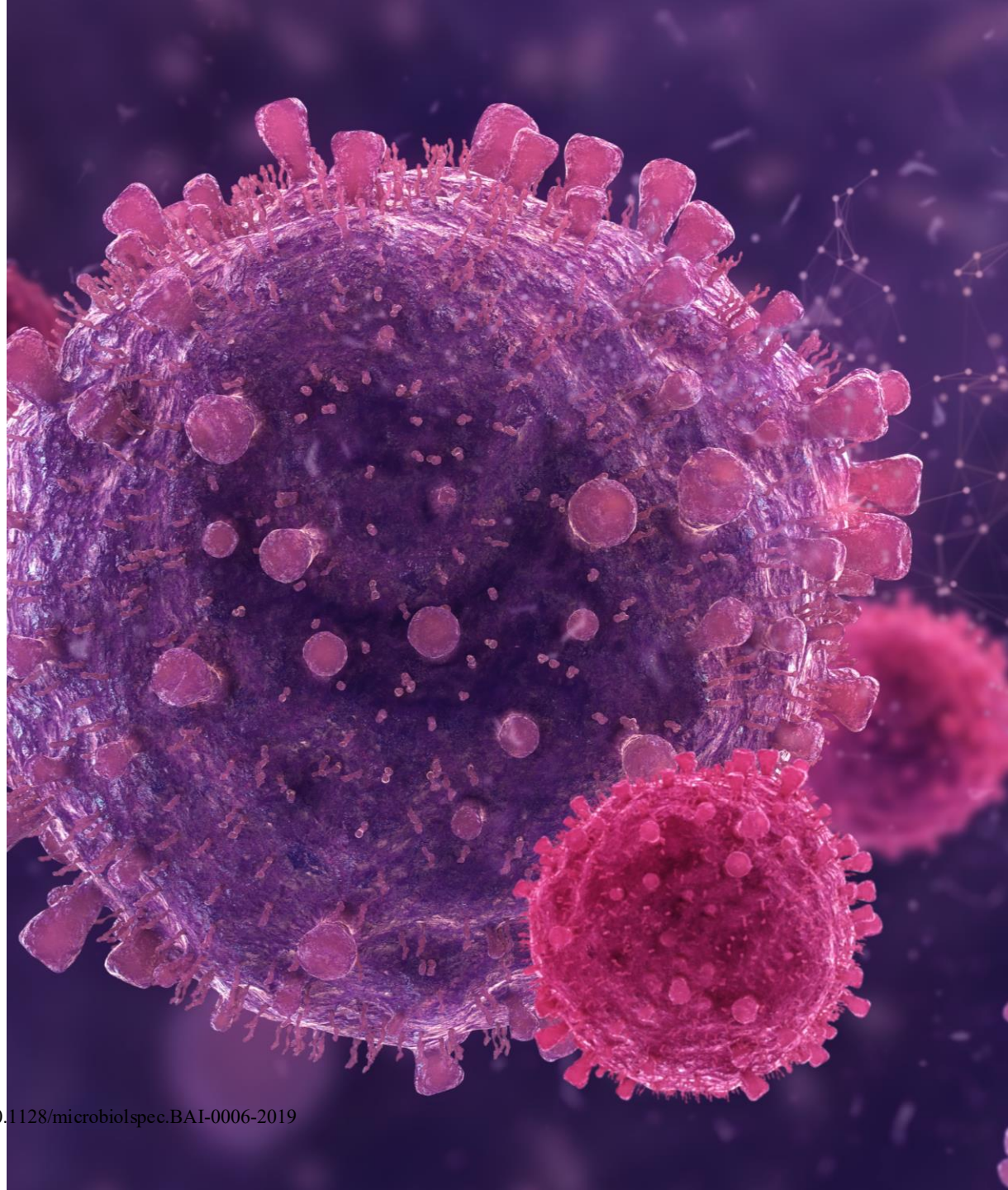
- Other transmission pathways:[9]
  - Consuming undercooked meat (sheep, cow, goat, or camel); game meat (bison, caribou, moose, elk, wild hog)
  - Consuming unpasteurized milk or other dairy products, like cheese
  - Inhaling bacteria (slaughterhouse)
  - Coming into contact with animal vaccines in the eyes, nose, or mouth or through injection while vaccinating animals



9. Centers for Disease Control and Prevention. About brucellosis. Brucellosis. May 2, 2024. Accessed April 9, 2025. https://www.cdc.gov/brucellosis/about/index.html#:~:text=Depending%20on%20the%20timing%20of,less%20than%202%25%20of%20cases

DESCRIPTION OF AGENT – LIFE CYCLE

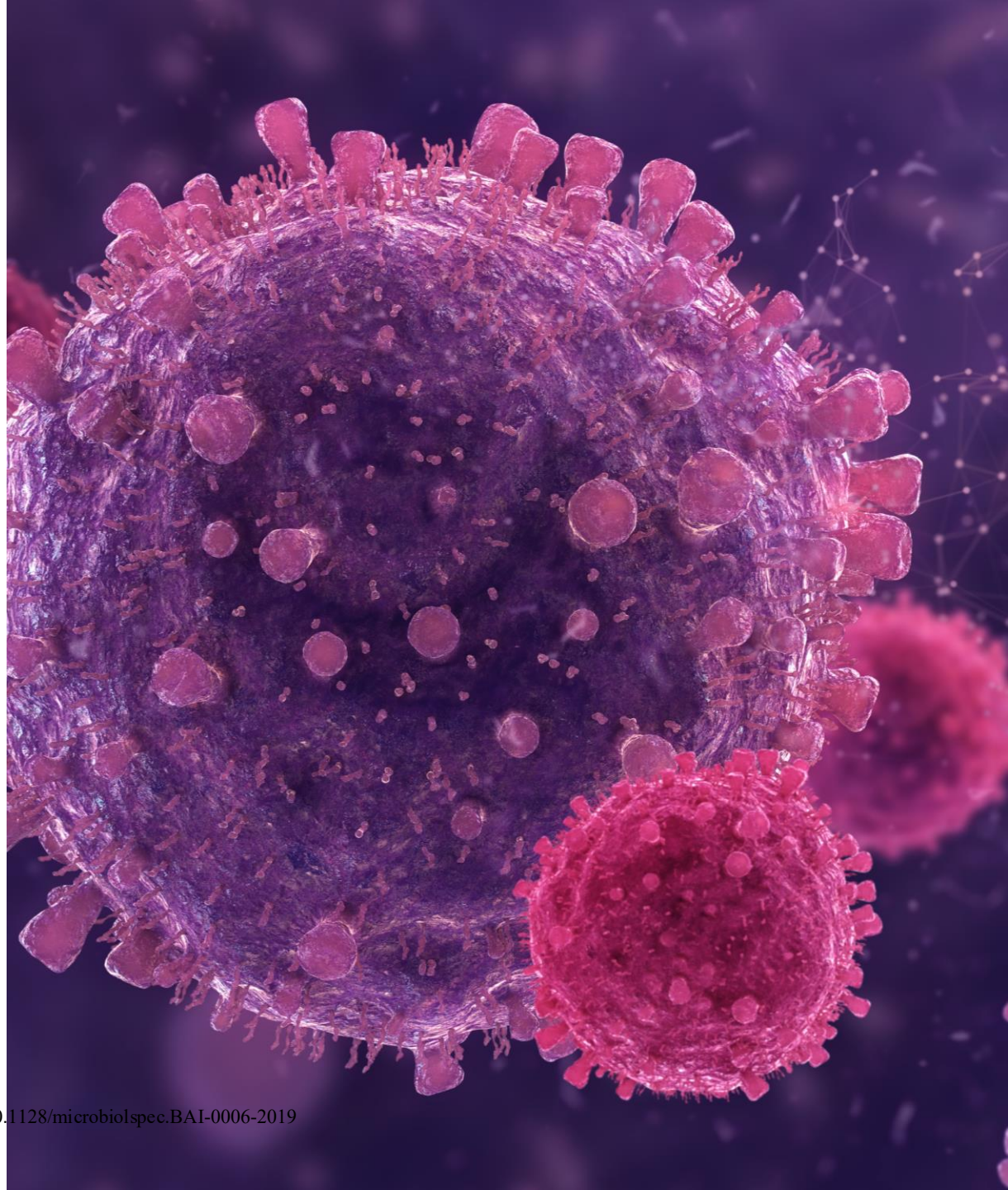
- Enters phagocytic or nonphagocytic cells[10]
- Becomes enclosed by membrane-bound compartment, termed the *Brucella*-containing vacuole (BCV), since the *Brucella* bacterium modifies the cells functionality into an idiosyncratic membrane-bound vacuole
- Travels along the endocytic pathway where it acquires early and late endosomal membrane markers --> pH = 4.5, similar to phagosomal maturation process
    - Termed: endosomal BCV (eBCV)
    - 0-8 hours postinfection



10. Celli J. The Intracellular Life Cycle of *Brucella* spp. *Microbiol Spectr*. 2019;7(2):10.1128/microbiolspec.bai-0006-2019. doi:10.1128/microbiolspec.BAI-0006-2019

DESCRIPTION OF AGENT – LIFE CYCLE CONT'D

- EBCV loses endosomal markers, interacts with the endoplasmic reticulum structures and gains ER membrane-associated markers; changes from the eBCV into an ER-derived organelle, seem to allow bacterial growth and replication or replicative BCVs (rBCVs)[10]
    - 8-12 hours postinfection
    - ER may interact with the Golgi apparatus, the host's secretory pathway
- Bacterial replication
    - 12-48 hours postinfection
- After bacterial replication, rBCVs are taken up by crescent-like membrane structures that create multimembrane vacuoles similar to autophagosomes that contain the bacteria and acquire late endosomal or lysosomal characteristics, autophagic BCVs (aBCVs)
    - 48-72 hours postinfection
    - Release bacteria to infect new cells



10. Celli J. The Intracellular Life Cycle of *Brucella* spp. *Microbiol Spectr*. 2019;7(2):10.1128/microbiolspec.bai-0006-2019. doi:10.1128/microbiolspec.BAI-0006-2019

BRUCELLOSIS SEVERITY - MORBIDITY

- Bosnia and Herzegovina[2]
- Turkey, Iran, Jordan, India, Pakistan
- China and Mongolia
- Sub-Saharan Africa: endemic; study from 2010-2019:
  - Human brucellosis: 0%-55.8% depending on regions
  - Cattle: 0.2%-43.8%
  - Goats: 0.0%-20.0%
  - Sheep: 0.0%-13.8%



2. Qureshi KA, Parvez A, Fahmy NA, et al. Brucellosis: epidemiology, pathogenesis, diagnosis and treatment-a comprehensive review. *Ann Med*. 2023;55(2):2295398. doi:10.1080/07853890.2023.2295398



BRUCELLOSIS SEVERITY - MORTALITY

- Less than 2% of cases; usually due to endocarditis[9]
  - Endocarditis from brucellosis very rare but accounts for most brucellosis-related deaths[11]
  - Brucella endocarditis needs rapid diagnosis and continued monitoring to improve treatment

9. Centers for Disease Control and Prevention. About brucellosis. Brucellosis. May 2, 2024. Accessed April 9, 2025. https://www.cdc.gov/brucellosis/about/index.html#:~:text=Depending%20on%20the%20timing%20of,less%20than%202%25%20of%20cases

BRUCELLOSIS PREVENTION[12]



Applying vast vaccination programs to decrease the prevalence



Identify infected animals, and eliminate them by slaughtering



Pasteurization of dairy products- avoid consumption of unpasteurized milk and dairy products, and also raw and undercooked animal products



Good hygiene and using protective clothing/equipment to prevent occupational exposures

12. Golshani M, Buozari S. A review of Brucellosis in Iran: Epidemiology, Risk Factors, Diagnosis, Control, and Prevention. *Iran Biomed J*. 2017;21(6):349-359. doi:10.18869/acadpub.ibj.21.6.349

BRUCELLOSIS TREATMENT



- Brucella spp. leads to chronic infections by destroying the immune system defenses, leading to fever, fatigue, and joint pain; sometimes endocarditis and neurological disorders[2]
- Diagnosis: clinical evaluations and laboratory examinations (blood cultures, serological testing, molecular testing)[2]
  - Diagnosis can be difficult from non-specific symptoms and lack of access to suitable samples for testing[2]
- Treatment: multiple antibiotics (some Brucella strains antibiotic-resistant) and vaccination of animals likely to be infected (especially livestock)[2]

2. Qureshi KA, Parvez A, Fahmy NA, et al. Brucellosis: epidemiology, pathogenesis, diagnosis and treatment-a comprehensive review. *Ann Med.* 2023;55(2):2295398. doi:10.1080/07853890.2023.2295398

HOST CHARACTERISTICS OF BRUCELLOSIS

Age: Adults, especially in working populations (farmers, veterinarians)[2]

Gender: More common in males due to occupation[13]

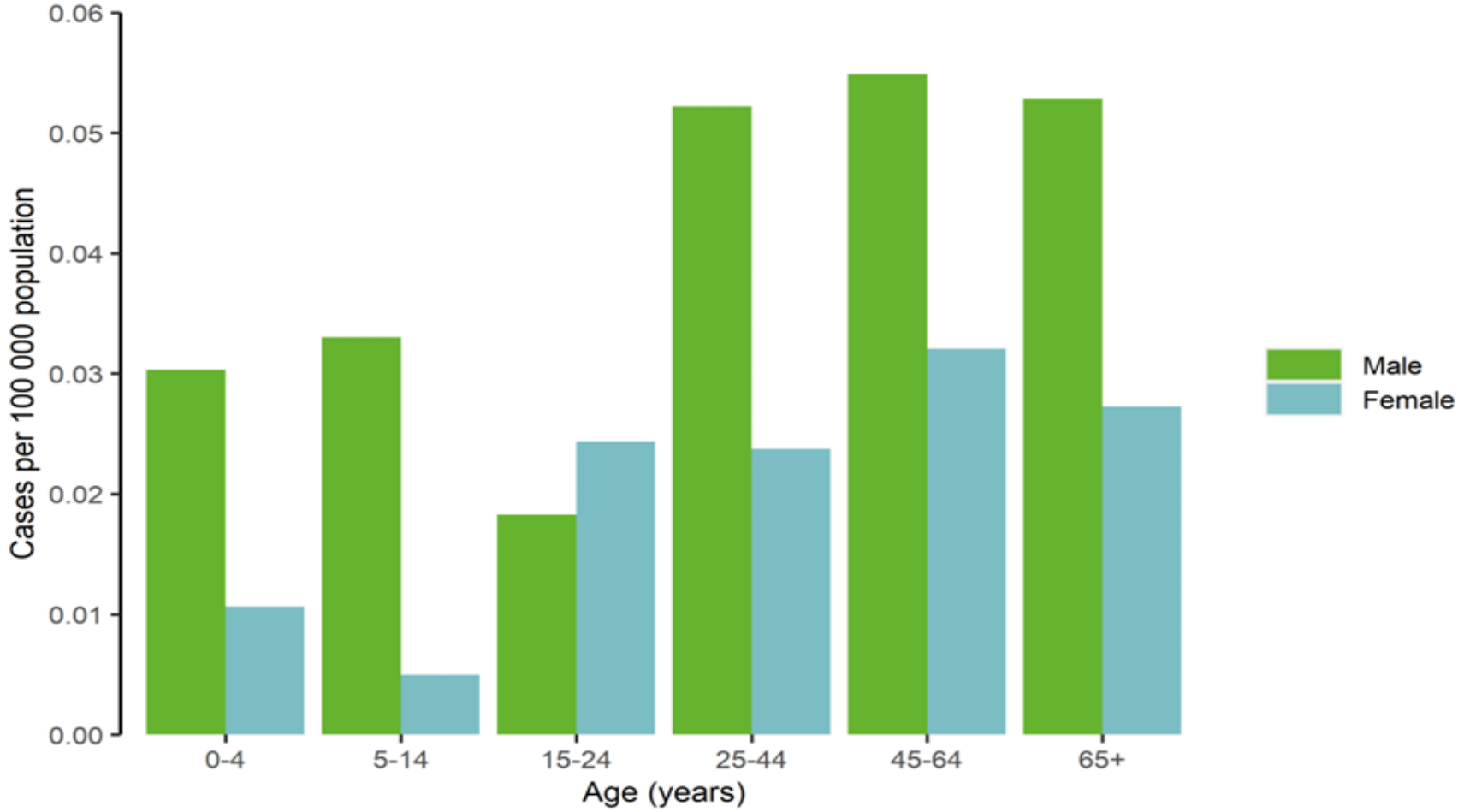
Nativity: Higher in regions where livestock farming is prevalent (e.g., Middle East, Central Asia)[2]

Marital Status: Higher in **married individuals** (due to livestock/domestic exposure)[2]

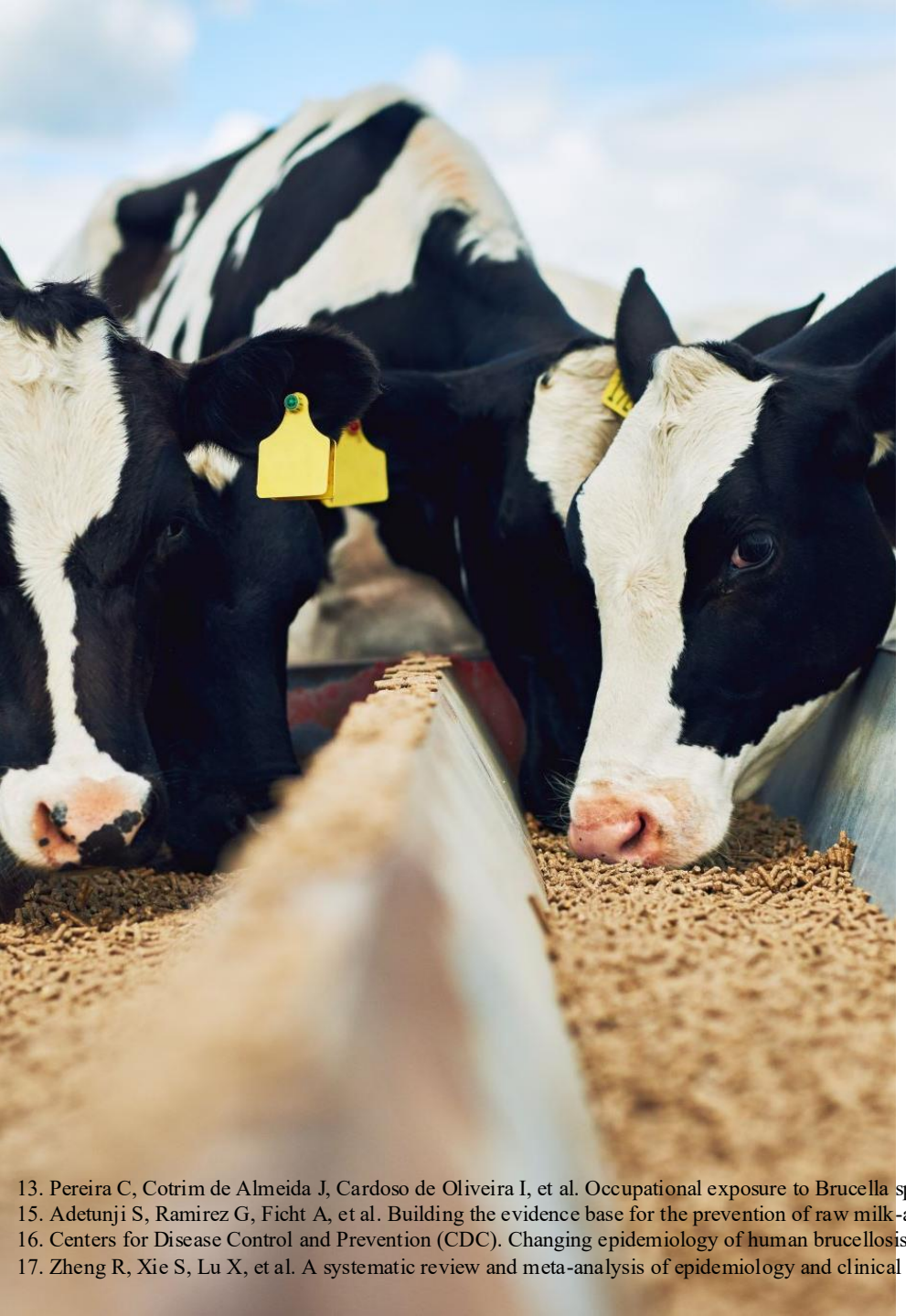
Family History/Genetics: Consider any genetic predispositions that may influence severity[14]

2. Qureshi KA, Parvez A, Fahmy NA, et al. Brucellosis: epidemiology, pathogenesis, diagnosis and treatment-a comprehensive review. *Ann Med*. 2023;55(2):2295398. doi:10.1080/07853890.2023.2295398
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HOST CHARACTERISTICS OF BRUCELLOSIS[3]



3. European Centre for Disease Prevention and Control. Brucellosis: Annual epidemiological report for 2022. Surveillance Report. April 2024. Accessed April 9, 2025. https://www.ecdc.europa.eu/sites/default/files/documents/BRUC_AER_2022_Report.pdf



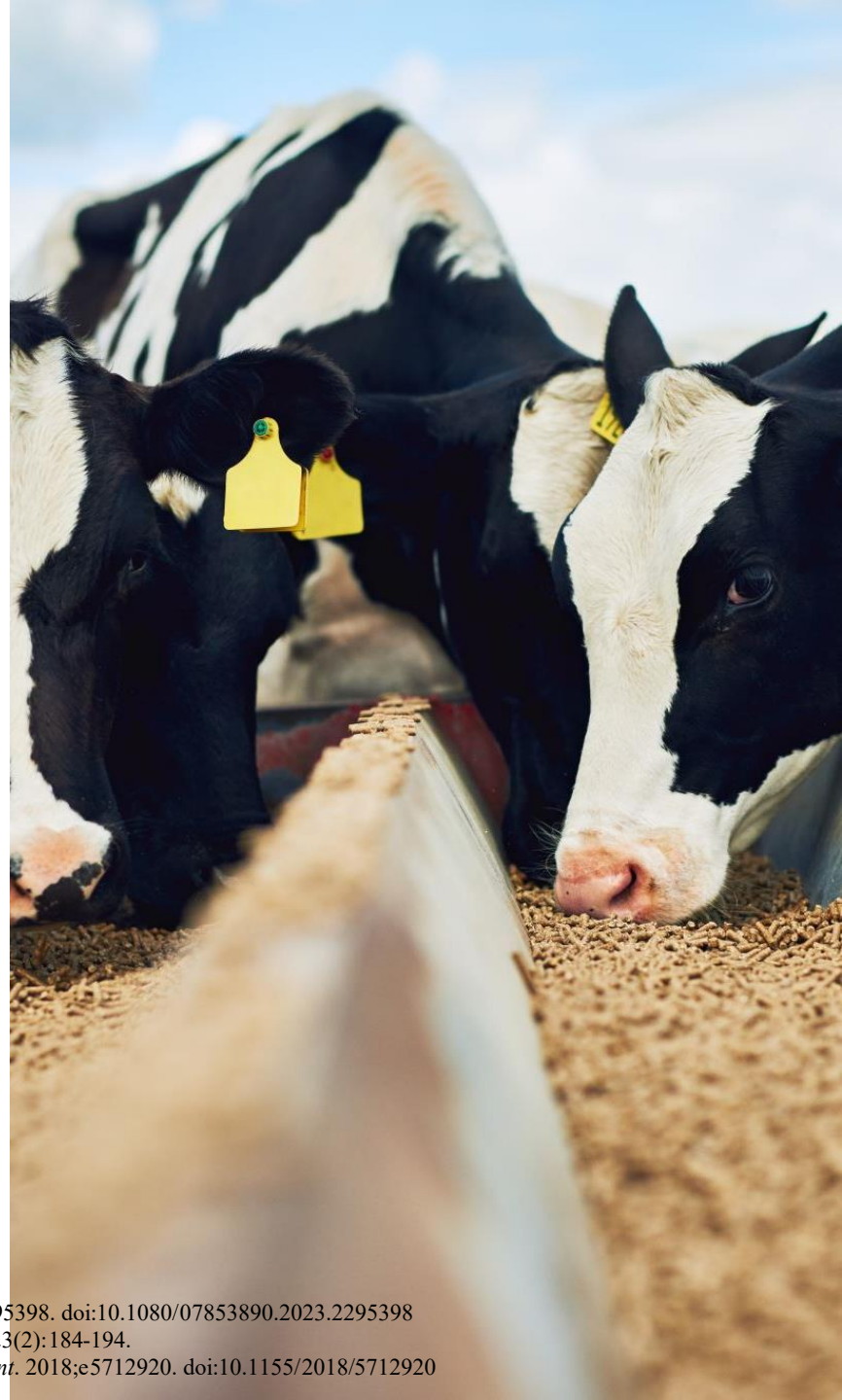
BRUCELLOSIS BEHAVIORAL CHARACTERISTICS

- **Risky Food Habits**[15]
- Eating unpasteurized dairy (raw milk, cheese, yogurt)
- Consuming undercooked meat from infected animals
- **Occupational Risks**[13]
- High-risk jobs: farmers, vets, abattoir workers, lab staff
- Lack of PPE increases exposure
- **Hygiene & Handling**[16,17]
- Poor hand hygiene after contact with animals/products
- Unsafe handling of birthing materials & aborted fetuses

13. Pereira C, Cotrim de Almeida J, Cardoso de Oliveira I, et al. Occupational exposure to Brucella spp.: A systematic review and meta-analysis. *PLoS Negl Trop Dis*. 2020;14(5): e0008164. doi:10.1371/journal.pntd.0008164
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16. Centers for Disease Control and Prevention (CDC). Changing epidemiology of human brucellosis, China, 1955-2014. *MMWR Morb Mortal Wkly Rep.* 2017;23(2):184-194.
17. Zheng R, Xie S, Lu X, et al. A systematic review and meta-analysis of epidemiology and clinical manifestations of human brucellosis in China. *Biomed Res Int*. 2018;e5712920. doi:10.1155/2018/5712920

BRUCELLOSIS BEHAVIORAL CHARACTERISTICS

- **Health-Seeking Behavior**[16]
- Delayed care and use of traditional remedies
- Non-adherence to long-term treatment
- **Cultural & Traditional Practices**[2]
- Preference for raw animal products
- Home slaughtering without hygiene standards
- **Education & Awareness**[16,17]
- Low knowledge of Brucellosis & safe practices
- Poor understanding of milk boiling, PPE, vaccination



2. Qureshi KA, Parvez A, Fahmy NA, et al. Brucellosis: epidemiology, pathogenesis, diagnosis and treatment-a comprehensive review. *Ann Med*. 2023;55(2):2295398. doi:10.1080/07853890.2023.2295398
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BRUCELLOSIS ENVIRONMENTAL ATTRIBUTES

- Geographic areas with high incidence (e.g., Mediterranean, Central Asia, Latin America)[3]
- Socioeconomic factors: poverty, limited access to healthcare
- Occupation: high-risk jobs related to animals[13]

3. European Centre for Disease Prevention and Control. Brucellosis: Annual epidemiological report for 2022. Surveillance Report. April 2024. Accessed April 9, 2025. https://www.ecdc.europa.eu/sites/default/files/documents/BRUC_AER_2022_Report.pdf
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TEMPORAL VARIATION OF BRUCELLOSIS[3]

Cyclical nature: higher incidence in rural farming seasons

Epidemics: occasional outbreaks following the spread of livestock

Secular trends: changes in prevalence over decades

3. European Centre for Disease Prevention and Control. Brucellosis: Annual epidemiological report for 2022. Surveillance Report. April 2024. Accessed April 9, 2025. https://www.ecdc.europa.eu/sites/default/files/documents/BRUC_AER_2022_Report.pdf

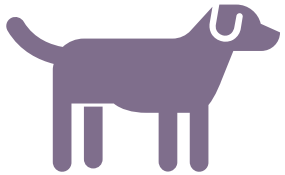
ADDITIONAL EPIDEMIOLOGIC VARIABLES[16,17]



Screening: serological tests for early detection



Vaccine: B. abortus vaccination in animals



Vector control (if applicable, though brucellosis primarily involves direct transmission)

16. Centers for Disease Control and Prevention (CDC). Changing epidemiology of human brucellosis, China, 1955-2014. *MMWR Morb Mortal Wkly Rep.* 2017;23(2):184-194.
17. Zheng R, Xie S, Lu X, et al. A systematic review and meta-analysis of epidemiology and clinical manifestations of human brucellosis in China. *Biomed Res Int*. 2018;e5712920. doi:10.1155/2018/5712920

CURRENT HYPOTHESES OF BRUCELLOSIS[16,17]

- Hypotheses on new transmission routes or emerging variants
- Research on vaccine efficacy and distribution strategies
- Potential zoonotic spillover dynamics

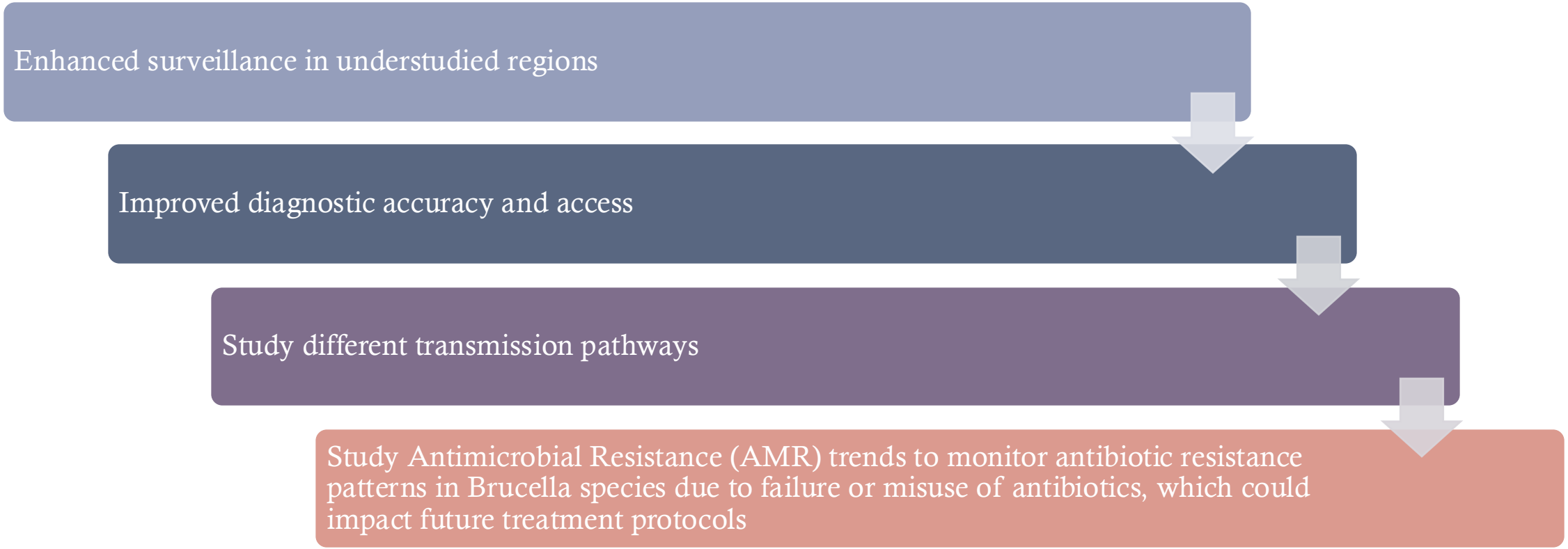
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GAPS IN KNOWLEDGE ON BRUCELLOSIS[17]

- Underreporting and misdiagnosis
- Limited Geographic Surveillance
- Lack of standardized reporting systems
- Transmission data gaps

17. Zheng R, Xie S, Lu X, et al. A systematic review and meta-analysis of epidemiology and clinical manifestations of human brucellosis in China. *Biomed Res Int*. 2018;e5712920. doi:10.1155/2018/5712920

SUGGESTIONS FOR FUTURE RESEARCH[17,18]



17. Zheng R, Xie S, Lu X, et al. A systematic review and meta-analysis of epidemiology and clinical manifestations of human brucellosis in China. *Biomed Res Int*. 2018;e5712920. doi:10.1155/2018/5712920
18. Harrison E, Prosada R. Brucellosis. *Pediatr. Rev*. 2018;39(4):222-224. doi:10.1542/pir.2017-0126

KEY POINTS

- Brucellosis is a neglected zoonotic disease with significant public health implications
- It is transmitted through unpasteurized dairy, direct animal contact, and occupational exposure
- Prevention focuses on animal vaccination, food safety, and protective measures for workers
- Treatment requires long-term antibiotics and early diagnosis to prevent complications
- Surveillance and control in endemic regions remain critical to reduce incidence

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11. Narimisa N, Razavi S, Khoshbayan A, et al. Prevalence of *Brucella* endocarditis: A systematic review and meta-analysis. *Health Sci. Rep*. 2023;6(5):e1301. doi:10.1002/hsr2.1301

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